Overview

Recent scholarly work (2020–2026) at the intersection of productive recovery, business resilience, and advanced technological approaches—specifically technology transfer, digital twins, and IoT—focuses strongly on post-disaster and seismic scenarios. The literature highlights both systematic reviews and practical case studies, emphasizing the integration of digital technologies in disaster risk management, urban planning, and resilience strategies. These studies are particularly attentive to the role of Industry 4.0, digital twins, and real-time data systems in enhancing infrastructure and community recovery after earthquakes.

• Industry 4.0 and Disaster Management

- Systematic reviews show the growing adoption of Industry 4.0 tools (big data analytics, IoT, robotics, cyber-physical systems) in disaster management and infrastructure resilience, mapping research activity to global disaster risk reduction frameworks [1].

- Bibliometric analyses identify gaps and prioritize future research directions, highlighting the synergy between digital transformation and resilient infrastructure.

• Digital Twin Technology for Urban and Seismic Resilience

- Critical assessments of digital twin applications in smart cities reveal their transformative role in urban planning and disaster response, especially for seismic resilience. Studies emphasize real-time simulation, data-driven decision-making, and alignment with sustainable development goals [2].

- Practical frameworks demonstrate how open-source digital twin models integrate multi-source data (geospatial, environmental, demographic) to support risk assessment, recovery coordination, and sustainable urban management in earthquake-prone regions [3].

Conclusion

The literature underscores that productive recovery and business resilience in post-disaster settings are increasingly supported by digital technologies and technological transfer. Digital twins, IoT, and Industry 4.0 concepts are central to both theoretical and practical advancements, enabling real-time, data-driven planning and decision-making for disaster risk reduction and post-earthquake recovery. These approaches not only enhance infrastructure resilience but also contribute toward sustainable cities and communities by bridging gaps between urban digitalization and disaster management. Integrating these technologies is seen as essential for achieving adaptive capacity, efficient recovery, and long-term sustainability in seismic and post-disaster contexts.

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1. Industry 4.0, Disaster Risk Management and Infrastructure Resilience: A Systematic Review and Bibliometric Analysis
Habibi Rad, M; Mojtahedi, M; Ostwald, M
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2. Exploring the Sustainability Benefits of Digital Twin Technology in Achieving Resilient Smart Cities During Strong Earthquake Events

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Habib, A; Habib, M; Bachir, H
2025 | ARABIAN JOURNAL FOR SCIENCE AND ENGINEERING

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3. A Digital Twin Approach to Sustainable Disaster Management: Case of Cayirova

Korkmaz, M; Akyildiz, Y; Ciftci, B
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